

PERSPECTIVE

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Transforming clinical medicine with multimodal artificial intelligence, agentic systems, and the model-context protocol: a perspective on future directions

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Abstract

This perspective explores the emerging integration of multimodal large language models (MLLMs), agentic artificial intelligence (AI) capabilities, and real-world healthcare infrastructure. Agentic AI refers to systems that can analyze and interpret complex data and autonomously perform goal-directed tasks, potentially transforming AI from passive tools into active participants in clinical workflows. To support such capabilities, new infrastructure is needed to enable secure and dynamic connectivity with diverse health information systems. Despite early progress, agentic AI faces major challenges, including limited interoperability across healthcare systems, underdeveloped tool ecosystems, and risks in autonomous decision-making. The recently proposed model-context protocol (MCP) aims to meet this need by offering a conceptual framework that could allow AI agents to retrieve, interpret, and act upon clinical data across environments such as electronic health records, imaging systems, and laboratory databases. We examine how MCP-style integration might enhance interoperability, contextual responsiveness, and safety in future medical AI applications. We also outline key limitations, including immature tool ecosystems, legal and ethical uncertainties, and computational constraints. While MCP does not yet constitute a widely adopted standard, it could provide a potential foundation for developing modular, reliable, and secure agentic systems. Looking ahead, the convergence of MLLMs, agentic AI, and MCP-style integration could enable more adaptive and collaborative AI systems in healthcare if key technical, ethical, and regulatory challenges are carefully addressed. This perspective synthesizes these concepts to propose a framework for integrating MLLMs, agentic AI, and MCP, outlining key applications and enhancing clinical interoperability.

Keywords Multimodal large language models, Agentic AI, Model-Context protocol, Medical AI integration, Healthcare



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1 Introduction

1.1 Emergence of MLLMs, agentic AI, and MCP

Recent advances in artificial intelligence (AI) are beginning to transform the landscape of modern healthcare [1]. Traditional machine learning and deep learning models in medicine have primarily focused on analyzing isolated data types, such as medical images or numerical parameters, and have often been limited to experimental settings or narrowly scoped commercial products. As a result, the integration of these models into real-world clinical workflows has remained limited [2]. However, the emergence of multimodal large language models (MLLMs) [3], agent-based AI systems [4], and integration frameworks such as the Model-Context Protocol (MCP) [5] has introduced a new paradigm for AI. These technologies could enable more flexible, context-aware, and automated systems that can support a broad range of clinical tasks [6, 7].

MLLMs such as GPT-4o, Med-PaLM 2, and Med-Gemini can interpret text, medical images, audio, and structured clinical data, aligning with the heterogeneous information flows of clinical care [8, 9]. Early real-world deployments indicate feasibility: a multicenter study by Mayo Clinic and UCSF used MLLMs to detect diagnostic delays in systemic mastocytosis from unstructured electronic health record (EHR) notes with reviewer-level performance across institutions [10]. Specialty-specific adaptations further show potential and constraints for clinical use, including the need for task-tuned models and careful evaluation in operational settings [11]. Beyond medicine, related agentic and integration approaches are emerging in finance, urban systems, and logistics, suggesting a broader shift from passive prediction to context-aware, goal-directed agents [12–14].

In parallel, the development of agentic AI has introduced the capability for AI systems to operate autonomously within defined constraints, enabling them to perform multi-step tasks, interact with external tools, and adapt dynamically to clinical workflows [15]. These developments require a foundation that supports secure and standardized communication between AI systems and clinical information sources. MCP addresses this need by enabling real-time access to relevant data and tools through a unified protocol. When combined with MLLMs and agentic AI, MCP could facilitate context-aware, interoperable, and tool-augmented intelligence that can be integrated into practical clinical systems.

1.2 Current clinical problem statement

Clinical information is dispersed across EHRs, imaging archives, laboratory systems, and communication tools, increasing cognitive load and leading to transcription errors and omissions during handoffs. Heterogeneous formats and interfaces hinder standard integration [16]. Current AI tools are typically single-purpose and weakly integrated; they lack provenance tracking, role- and consent-aware access, mechanisms for human oversight, and predictable failure behavior, which diminishes trust and accountability. As a result, clinicians must spend additional time synthesizing outputs before action. A standards-aligned, modular architecture that enables supervised and auditable agent workflows, under explicit governance with clearly defined validation, is required for safe and effective deployment in clinical care.

1.3 Scope and purpose of this perspective

This perspective explores how the convergence of MLLMs, agentic AI, and integration protocols such as MCP-style approaches may shape the next generation of clinical AI systems. This article highlights selected developments that illustrate a shift from isolated predictive models to context-aware and tool-integrated medical AI. Rather than presenting a comprehensive review, it presents representative applications across diagnostics, decision support, documentation, and patient interaction, while also addressing limitations related to implementation, interoperability, data privacy, and regulatory considerations. As a perspective piece, it does not provide empirical validation or exhaustive literature coverage, and its conclusions should be interpreted as conceptual guidance rather than definitive evidence. By synthesizing current progress and identifying key challenges, this perspective aims to encourage critical dialogue on the evolving role of AI in healthcare and to explore how MLLMs, agentic frameworks, and MCP-based infrastructures might be responsibly incorporated into clinical workflows (Fig. 1). We focus on healthcare because it couples heterogeneous data sources with stringent privacy obligations and time-critical decisions, which together make reliability, safety, and interoperability paramount.

We conducted a targeted narrative review to contextualize multimodal AI, agentic systems, and the MCP in healthcare. Sources included PubMed/MEDLINE, Scopus, arXiv, reputable news articles, and selected conference proceedings. Search strings combined method and domain terms. Example: (“multimodal large language model” OR “vision language model” OR “agentic artificial intelligence” OR “model context protocol”) AND (“clinical” OR “electronic health record” OR “medical imaging” OR “interoperability” OR “medicine”). The window covered publications and preprints from January 2022 to March 2025. We prioritized peer-reviewed studies, technical frameworks, and large pilot reports that described clinical integration, evaluation, or governance. Titles and abstracts were screened, duplicates were removed, and potentially relevant full texts

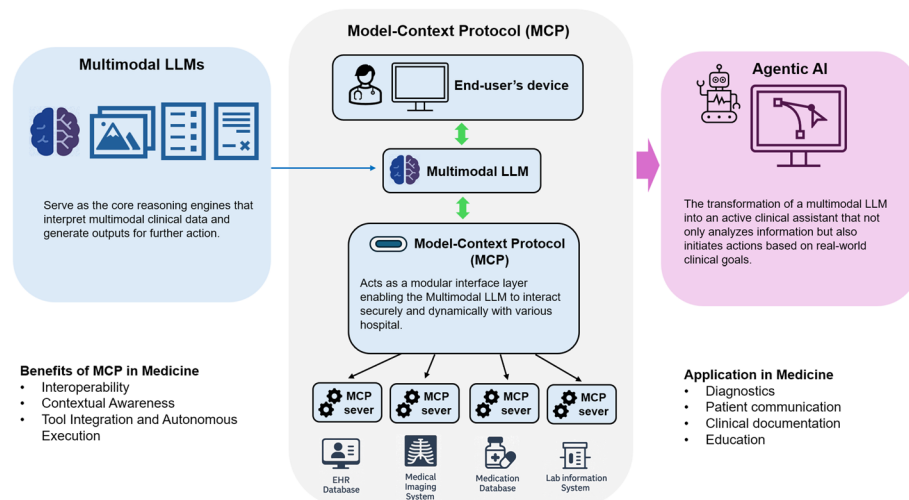


Fig. 1 Conceptual framework of the model-context protocol (MCP) enabling agentic AI in medicine. This figure illustrates the integration of Multimodal Large Language Models (multimodal LLMs), the MCP, and Agentic AI systems in clinical environments. The MCP acts as a modular interface layer, allowing the MLLM to dynamically and securely interact with external clinical systems such as EHR databases, medical imaging archives, medication information systems, and laboratory databases. Through MCP-facilitated tool integration and real-time contextual awareness, MLLMs can evolve into Agentic AI systems that not only analyze information, but also autonomously initiate actions aligned with clinical goals

were reviewed for inclusion. This article is a perspective, not a systematic review. Selection emphasized representative capabilities, risks, and infrastructure discussions rather than exhaustiveness.

2 Key concepts and current status

2.1 Multimodal large Language models (MLLMs)

MLLMs are advanced AI systems capable of processing and generating outputs from multiple types of input data [17]. While traditional language models, such as early versions of GPT, demonstrate strong performance in text-based reasoning, they are limited in their ability to interpret non-textual data such as images, audio, and other sensor-derived signals. In clinical settings, however, essential information is distributed across a wide range of modalities, including narrative documentation, medical imaging (e.g., X-rays, MRIs, fundus photographs, and pathology slides), time-series data (such as vital signs and electrocardiograms), audio recordings (e.g., heart or lung sounds, patient interviews), video streams (e.g., surgical procedures), and high-dimensional data such as genomic or proteomic profiles [18]. MLLMs are designed to integrate these heterogeneous data types within a single computational framework, enabling joint analysis and reasoning that more closely reflects the complexity of real-world clinical decision-making. For instance, an MLLM may simultaneously evaluate a radiographic image alongside a pathology report using image encoders and optical character recognition where appropriate, thereby generating a more comprehensive interpretation than would be possible with text or image alone [19]. In ophthalmology, this approach could support the integration of ocular biometry and anterior segment imaging to enhance surgical planning and risk stratification [20].

Recent research has increasingly emphasized the importance of MLLMs in the medical field. Although unimodal language models based solely on text demonstrate strong reasoning capabilities, a previous article noted that they do not fully capture the complex and varied nature of clinical practice, highlighting the need for multimodal integration [3]. In response, several research groups have begun developing MLLMs tailored to healthcare applications. For instance, a medical MLLM designed for radiologic analysis was reported to learn extensive clinical knowledge by processing both X-ray images and corresponding textual reports [21]. In the general domain, models like GPT-4o or Gemini 2.5 have incorporated visual processing alongside text, an approach now being adapted for healthcare. Large-scale vision-language models such as LLaVA-Med, PMC-VQA, and Med-Flamingo are examples of general-purpose MLLMs fine-tuned for medical imaging tasks [22]. Overall, MLLMs are positioned as a foundational technology for advanced medical AI, offering the capacity to interpret diverse data sources and support diagnostic and analytical processes in complex clinical environments.

2.2 Agentic AI

While large language models have been widely adopted as advanced text-based reasoning engines that generate responses to user prompts, agentic AI systems are characterized by their ability to act autonomously and pursue goal-oriented tasks [4]. Unlike conventional AI models that merely provide answers to questions, agentic AI is designed to take proactive actions, make decisions, and interact with external tools or digital environments to achieve specified objectives. These systems operate through an

internal cycle of observation, reasoning, and action, enabling them to coordinate multi-step processes and adapt dynamically, rather than being limited to single-turn interactions. Reflection allows models to evaluate their own outputs and engage in iterative self-correction. Planning facilitates structured task decomposition, while multi-agent collaboration supports the division of labor among specialized models, each handling a component of a larger task. Recent advances in MLLMs have significantly accelerated the development of agentic AI [23].

Formally, agentic AI refers to intelligent architectures capable of deploying autonomous agents that not only generate language outputs but also perform actions, retrieve data, and orchestrate processes to fulfill high-level instructions [4, 24]. In healthcare, this capability will be particularly impactful [15]. For example, an agentic AI system could autonomously retrieve a patient's laboratory results and imaging, summarize the findings for clinical review, initiate a draft order for additional testing based on abnormalities, and notify the care team. These agents may perceive their environment through integration with clinical systems, reason about the significance of available information, and take targeted actions to advance their assigned goals. General-purpose implementations such as AutoGPT and HuggingGPT have already demonstrated the feasibility of chaining together reasoning steps with tool execution in open-ended problem solving [25]. The same underlying principle applies to medicine, where agents might query EHRs, consult medical knowledge bases, and utilize diagnostic interfaces to support clinical decisions. Crucially, the intent of agentic AI in healthcare is to augment, not replace, human clinicians. By automating routine yet complex administrative or analytical tasks, these systems can relieve clinicians of cognitive and logistical burdens, allowing them to focus on patient-centered care and complex decision-making.

2.3 Model-context protocol (MCP-style integration)

Deploying AI in clinical settings depends on robust, scalable interoperability with a heterogeneous digital infrastructure, including EHRs, imaging archives, laboratory information systems, scheduling platforms, and medication or decision-support databases. Historically, integrations have used bespoke point-to-point links between models and data sources. These ad hoc solutions are difficult to generalize, costly to maintain, and ill-suited to the dynamic and heterogeneous nature of healthcare data. The MCP, introduced by Anthropic in 2024, seeks to address these challenges by providing an open, standardized mechanism for secure tool and data access [5]. MCP defines a universal interface between AI agents and real-world systems, enabling modular and reusable connections that support dynamic, real-time information retrieval and task execution via MLLMs. An apt comparison is to the USB-C standard in hardware design [26]. Just as USB-C enables a wide variety of peripheral devices, such as displays, storage drives, and power supplies, to interface with a host system through a unified connector, MCP allows AI agents to interact with a wide range of clinical systems through a consistent protocol. Each MCP server could expose a particular set of capabilities, such as laboratory data, imaging repositories, or medication databases. The MCP client, typically an AI assistant powered by an MLLM, could query and interact with these resources without requiring system-specific logic (Fig. 2).

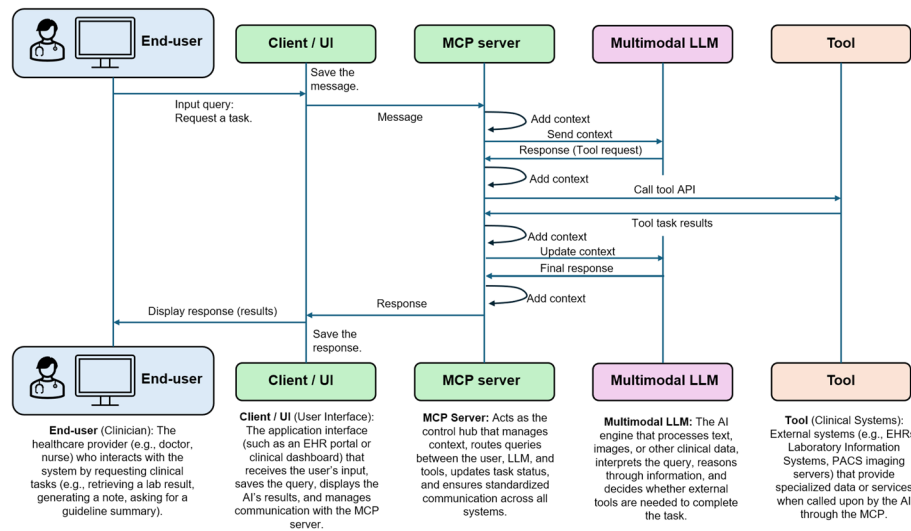


Fig. 2 Workflow of AI systems using the model-context protocol (MCP). This diagram illustrates the interaction flow between end-users (clinicians), the client user interface (UI), MCP server, multimodal large language models (Multimodal LLMs), and external clinical tools

Table 1 Expected benefits of applying MCP-style integration in medical AI systems

Classification	General benefit	Clinicians	Patients	Concerns
Interoperability	Standardized communication protocol that connects AI systems to diverse data sources and tools	Reduced integration burden; easier to scale AI across systems without custom interfaces	Improved care continuity and system-wide coordination of services	- Requires industry adoption of MCP standards - Key and certificate management errors - Limited tool support today
Contextual awareness	Real-time retrieval and updating of patient-specific information for adaptive AI behavior	Up-to-date insights at the point of care; supports more informed and timely decisions	More accurate and personalized recommendations based on real-time data	- Risk of privacy breaches if context is not managed with strict access control - Conflicting sources causing wrong context/provenance gaps that hinder audit and correction
Tool integration and autonomous execution	AI agents can execute clinical tasks by interacting with external systems dynamically	Automates repetitive tasks, supports workflow orchestration, and improves efficiency	Faster care delivery, fewer delays, and better safety through automated checks	- Unsafe tool invocation or race conditions - Prompt injection and indirect data exfiltration through tools - Liability for semi-autonomous actions - Alert fatigue and overreliance that reduces human oversight - Vulnerability to multimodal adversarial attacks that combine text, images, and structured inputs

Model Context Protocol (MCP)

MCP-style integration is expected to be essential for creating agentic AI that actively pursues clinical goals rather than merely analyzing data. The benefits of MCP in medical AI systems are significant (Table 1):

- **Interoperability:** MCP-style integration could replace fragmented, hard-coded connections with a standardized communication protocol. Hospitals and software developers can expand AI functionality by simply deploying additional MCP servers, without modifying the core behavior of the AI system. As the ecosystem of MCP-compatible components grows, AI tools can interact with a broad range of clinical

and biomedical data sources. This promotes broader evidence-based reasoning and a more integrated clinical decision-support experience.

- **Real-time contextual awareness:** MCP-style integration could allow AI systems to retrieve patient-specific context in real time, rather than relying solely on pre-trained knowledge or static prompts. During clinical use, an AI assistant can query the latest lab results, integrate them with existing notes, and reference relevant clinical guidelines, all through structured MCP interactions. This capability allows the AI to maintain situational awareness and generate responses that are both accurate and patient-specific.
- **Tool integration and autonomous execution:** MCP-style integration could transform external services, such as application programming interfaces (APIs) and clinical applications, into callable resources for AI agents. This functionality supports agentic behavior, where the AI is able to act based on its internal reasoning. For instance, an AI assistant may access a patient's medication history through one MCP interface and then check for potential interactions using another, automatically guiding its recommendations. Because these tools are accessed through a unified protocol, the AI system does not require hard-coded instructions for each system and can interact with them dynamically.

Early adoption of MCP has already been observed in software engineering and enterprise domains, with platforms such as Replit and Microsoft's Copilot Studio leveraging the protocol for tool-augmented reasoning in non-clinical settings [5]. We assume an MCP-style interface for auditable tool calls; current implementations vary, with evolving protocol versions and community adapters (e.g., LangChain) that ease client integration but do not eliminate the need for institution-specific security and observability. Similar applications in medicine are anticipated. For example, EHR vendors may expose structured patient data through MCP endpoints, and medical devices may stream real-time physiological signals directly to MCP-compatible agents. In summary, MCP can provide a foundational interoperability layer that enables context-aware, secure, and scalable integration of advanced AI into clinical workflows.

2.4 State of the art: current models in medical AI

Building on MLLMs, agentic design patterns, and emerging integration protocols, several models and platforms now exhibit capabilities that are necessary for future agentic behavior, although fully autonomous, clinically deployed agents are not yet realized. Common enabling features include tool-use APIs, multimodal inputs, long-context processing, and task-specific interfaces that support interactive, context-aware workflows.

- **GPT family and optimized variants:** OpenAI's GPT-4, released in 2023, represents one of the most advanced MLLMs currently available. It accepts both text and image inputs, making it a fully multimodal model, and has demonstrated human-level performance across various standardized assessments [20]. In the medical domain, GPT-4 has been shown to pass the United States Medical Licensing Examination (USMLE) and similar assessments, consistently exceeding passing thresholds [27]. In comparative studies, GPT-4o (a more optimized variant of GPT-4 with enhanced capabilities) achieved high accuracy on the USMLE and comparable scores on

international medical exams, outperforming both previous models and many human examinees [28]. Beyond standardized testing, GPT-4 and its derivatives have been applied to detailed clinical tasks across diverse settings [6], where they have exhibited robust performance in supporting medical reasoning and documentation workflows [29]. They can function within clinical calculators and analytics pipelines, interpreting structured variables and free text to generate semantically guided decision support [30, 31]. GPT-4o also includes functionality for agentic tasks via OpenAI's function-calling API and plug-in ecosystem, allowing it to retrieve data, interface with external services, and perform structured tasks.

- **Med-PaLM 2:** Med-PaLM 2, developed by Google Research and DeepMind, is a specialized MLLM fine-tuned on medical texts and expert demonstrations. Based on the PaLM-2 architecture, Med-PaLM 2 achieved 86.5% accuracy on the MedQA benchmark, significantly surpassing earlier models [32]. In clinical trials at institutions such as the Mayo Clinic, Med-PaLM 2 was piloted to support physician Q&A tasks. While some limitations in factual consistency remain, early evaluations indicate performance comparable to that of practicing physicians on several tasks [33]. The model supports long-form responses and demonstrates improved alignment with clinical reasoning due to its fine-tuning approach.
- **Gemini and Med-Gemini:** DeepMind's Gemini represents Google's largest and most comprehensive multimodal AI system, developed with native support for planning, reasoning, and tool use. Med-Gemini is a healthcare-specific extension of Gemini that integrates text, image, 3D, and genomic data capabilities. According to Google, Med-Gemini achieved 91.1% accuracy on the MedQA benchmark and outperformed competing models on multiple clinical reasoning tasks [34]. It has also been evaluated for medical summarization and correspondence, where clinicians rated its output as equivalent to or better than human-written text in approximately 50% of cases. Sub-models of Med-Gemini include Med-Gemini-3D, capable of interpreting radiologic images and generating diagnostic reports, and Med-Gemini-Polygenic, which analyzes genomic data to infer risk profiles [35].
- **Other notable platforms and models:** Claude emphasizes extended context and safety features for large-document processing in healthcare workflows [36, 37]. Purpose-built clinical models (for example, Hippocratic AI) focus on patient-facing use cases and safety benchmarking [38]. Open-source biomedical large LLMs such as BioMedLM, BioGPT, and PMC-LLaMA support privacy-sensitive deployments and literature-centric tasks [39]. In vision, domain-adapted foundation models (for example, VisionFM) show cross-modality generalization for ophthalmic imaging and related diagnostic tasks [40]. DeepSeek has recently been benchmarked against other popular MLLMs on medical tasks, showing USMLE accuracy of 0.92 (slightly below ChatGPT-o1 at 0.95 and above Llama 3.1-405B at 0.83), similar performance on text-based diagnostic and management cases and on RECIST 1.1 tumor classification, higher rated diagnostic reasoning steps, and lower quality for summarized imaging reports compared with ChatGPT-o1 [41].

Current systems do not yet meet the bar for fully agentic clinical AI, but many incorporate capabilities that enable agent-like behavior, particularly multimodality, tool

access, and long-context reasoning. Near-term utility concentrates on documentation assistance, information retrieval, and structured triage.

2.5 Applications in medicine

The combination of multimodal AI models, agentic capabilities, and integrative protocols is enabling a wide range of applications in healthcare [42]. In this section, we explore current and emerging uses in key areas: diagnostics, clinical decision support, documentation, medical education, and patient interaction. Many of these applications are already in pilot programs or early deployment, offering a glimpse of how routine medicine might change in the future.

- **Preliminary application of agentic AI:** Table 2 summarizes early, real-world deployments that exhibit agent-like behavior within constrained tasks. In decision support, a Microsoft system demonstrated sequential diagnostic reasoning by querying symptoms, ordering pertinent tests, and iteratively refining hypotheses, outperforming single-turn MLLMs in simulated cases [43]. In documentation, voice assistants such as Suki AI show semi-agentic behavior (automatic transcription, retrieval, and structured note generation) with reported reductions in documentation time and improved note completeness and satisfaction [44]. Patient-facing preoperative platforms developed with Nvidia and The Ottawa Hospital deliver multimodal education and support, reducing in-person assessments and staff hours in reported evaluations [45]. On the administrative side, VoiceCare AI's prior-authorization agent and Ushur's conversational systems have offloaded large volumes of payer and service-request interactions and improved turnaround times in operational settings [46, 47]. In the physical domain, Diligent Robotics' Moxi autonomously navigates hospitals to deliver medications and supplies, decreasing manual walking distance and staff workload [48]. Most systems remain semi-agentic, bounded in scope with explicit human oversight, but they demonstrate measurable workflow gains and inform governance needs for broader adoption.
- **Diagnostics and clinical decision support:** AI is increasingly used for disease identification, test selection, and treatment planning. Vision-capable MLLMs interpret imaging studies (for example, radiography and MRI), draft preliminary reports, and triage findings; related advances in digital pathology and dermatology support early malignancy detection [49]. Unlike stand-alone imaging models, MLLMs integrate language and vision, enabling interactive dialogue that conditions interpretations on symptoms and history [50]. Emerging agentic patterns allow systems to retrieve clarifying information from clinical databases or knowledge resources and to orchestrate domain tools, moving beyond static decision trees toward adaptive, context-aware support [51, 52]. Treatment recommendation remains nascent but is trending toward guideline-concordant, patient-specific proposals grounded in clinical history and, where available, genomics.
- **Clinical documentation and data management:** Documentation is a leading area of impact for LLMs and agentic systems [53]. Tools such as Nuance DAX Express combine speech recognition with language understanding to generate structured notes from encounters, reducing documentation burden and allowing clinicians to redirect time to patient care [54]. Long-context models can summarize fragmented

Table 2 Preliminary applications incorporating agentic features in healthcare: current systems and functional classification

Application domain	Description	Implementing organizations	Reported outcomes	Agentic classification
Clinical Decision Support via Sequential Diagnosis	LLM agent performs sequential diagnostic reasoning by querying symptoms, ordering tests, and refining hypotheses	Microsoft AI [43]	Improved diagnostic accuracy and efficiency vs. single-shot LLMs in simulated patient cases	Agentic: Goal-directed reasoning with dynamic action selection and patient context integration
Clinical Documentation Automation	AI-driven voice assistants that autonomously transcribe physician–patient interactions, retrieve clinical information, generate structured documentation, and initiate follow-up tasks	Suki AI, integrated into major electronic health record (EHR) systems (e.g., Epic, Cerner); deployed in institutions including Rush University Medical Center and MedStar Health [44]	Documentation time reduced by up to 72%; improved note completeness and quality; increased physician satisfaction and reduced after-hours workload	Semi-Agentic: Partial task autonomy and limited tool invocation without full context reasoning or goal decomposition
Virtual Preoperative Patient Engagement	AI-enabled digital avatars providing personalized preoperative instructions, procedural education, and continuous patient support through multi-modal communication channels	Collaborative development by Nvidia and The Ottawa Hospital (Canada) [45]	Reduction in duration of in-person preoperative assessments by approximately 50%; enhanced patient comprehension and preparedness; estimated saving of approximately 80,000 clinical staff hours annually	Semi-Agentic: Automated interaction with personalization, but lacking dynamic task management
Autonomous Prior Authorization and Claims Processing	AI agents conducting insurance verification, submission of prior authorization requests, and management of follow-up communications with payers, thereby replacing manual administrative workflows	VoiceCare AI (“Joy” agent), piloted at Mayo Clinic and deployed across imaging and surgical services [46]	Offloaded approximately 70,000 insurance-related telephone interactions per month; improved authorization turnaround time; enabled reallocation of administrative staff to higher-value tasks	Semi-Agentic: Operational autonomy in structured tasks, limited clinical context reasoning
Automated Member Services in Health Insurance	Conversational AI platforms autonomously addressing insurance member service requests, including policy updates, card reissuance, and appointment scheduling, without human agent intervention	Ushur, operationalized within major U.S. health insurance providers (specific organizations undisclosed) [47]	Resolution of over 36,000 service requests within two months; projected annual operational savings of approximately USD 14 million; enhanced member satisfaction through immediate response capabilities	Semi-Agentic: High conversational autonomy, but fixed workflows and no clinical action
Hospital Logistics and Autonomous Delivery Systems	Mobile robotic platforms facilitating autonomous transport of medications, specimens, and supplies within hospital environments, including navigation of elevators and secured areas	Diligent Robotics (“Moxi”), deployed at institutions such as Northwestern Medicine, Cedars-Sinai Medical Center, and MultiCare Health System across over 100 healthcare facilities [48]	Substantial reduction in manual delivery tasks; at Cedars-Sinai, two robotic units collectively eliminated nearly 300 miles of manual walking distance over six weeks; contributed to improved clinical efficiency and decreased staff physical burden	Agentic (Physical Task Only): Autonomous navigation and execution, though not reasoning over clinical data

records, retrieve salient details across documents, and assemble longitudinal narratives that support decision making while reducing oversight risk [55–57]. Agentic capabilities further extend to coding support, form completion, and claims assistance by extracting required data and interfacing with institutional workflows. As these systems mature [58], a clinician-in-the-loop assistant that manages routine documentation and administrative tasks, with auditable provenance and predictable failure behavior, appears increasingly feasible.

- **Medical education and training:** AI, particularly through MLLMs and agentic AI, is increasingly influencing medical education across all levels [59]. For students and trainees, MLLMs can function as on-demand tutors, capable of explaining complex medical concepts, generating practice questions, and providing real-time feedback during virtual case simulations [60, 61]. These models not only answer queries but also elucidate the underlying reasoning [62], offering an interactive and low-stakes learning environment that supplements traditional education. Given their demonstrated performance exceeding that of many medical students on standardized assessments, LLMs show promise in creating high-quality educational resources. Similarly, practicing clinicians benefit from AI-powered tools that summarize emerging research, clinical guidelines, and specialty-specific updates, facilitating continuing education and supporting evidence-based decision-making during patient care.
- **Patient interaction and engagement:** Generative systems translate clinical information into accessible language, draft after-visit summaries, and support empathetic, personalized communication [63–65]. Modern MLLMs power conversational symptom checkers and triage tools that clarify concerns and direct patients to appropriate care levels; when integrated with scheduling, semi-agentic workflows can also facilitate appointment booking [66, 67]. Pre-visit data collection improves telehealth efficiency by capturing structured histories in advance [68]. Within patient portals, assistants help clinicians by drafting message replies and summarizing charts, while voice-based home agents answer health questions and integrate device data for disease monitoring [69]. Mental-health chatbots extend these capabilities with cognitive-behavioral strategies and supportive dialogue as adjuncts to care [70].

Early applications concentrate on documentation, information retrieval, structured triage, and narrowly scoped decision support. Most systems operate under human oversight with bounded autonomy. Progress toward broader, goal-directed use will depend on advances in calibration, interpretability, safety alignment, and rigorous evaluation, alongside interoperable, auditable integration with clinical systems.

3 MCP and the future of agentic medical AI systems

3.1 MCP: enabling scalable integration of agentic AI

The integration of MLLMs with agentic capabilities into real-world clinical workflows requires seamless, scalable interaction with diverse healthcare systems. The MCP provides a standardized foundation for achieving this integration. By enabling secure, dynamic access to EHR, imaging archives, laboratory information systems, and scheduling platforms, MCP allows AI agents to interact with heterogeneous clinical

environments without the need for bespoke interfaces (Fig. 3). Instead of system-specific APIs, agents issue standardized calls for patient data and services, improving interoperability and accelerating deployment.

Beyond static retrieval, MCP-style integration could support on-demand context augmentation during clinical interactions [5]. Rather than preloading all relevant information into a prompt, an agent can query connected databases in real time based on conversational needs. For example, an agent could use MCP to fetch prior colonoscopy reports, align findings with guideline logic, and assemble a tailored summary; while such end-to-end autonomy remains exploratory, this pattern illustrates how MCP extends an agent's effective context beyond model memory to live clinical systems. MCP also enables action orchestration under clinician oversight: identifying missing tests, drafting laboratory orders, or initiating scheduling via standardized functions. Clinical approval remains essential, but MCP reduces technical friction by providing consistent, auditable operations across vendors and sites. In multi-tool settings, MCP promotes context consistency by ensuring that agents draw from unified sources, mitigating discrepancies introduced by fragmented integrations.

Security and auditability should be first-class properties in an MCP-based architecture. All exchanges can be logged and encrypted, with centralized access controls that enforce role- and consent-aware policies, least-privilege permissions, and just-in-time access. Comprehensive audit trails should record capability invocations, inputs, outputs, and acting identities to support monitoring, incident response, and quality review. These controls must align with current and evolving privacy regulations, including the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR), and be complemented by data minimization, key management, and retention policies [71]. Governance frameworks can further restrict agent access to approved datasets and actions, thereby strengthening institutional oversight of AI activity in clinical settings.

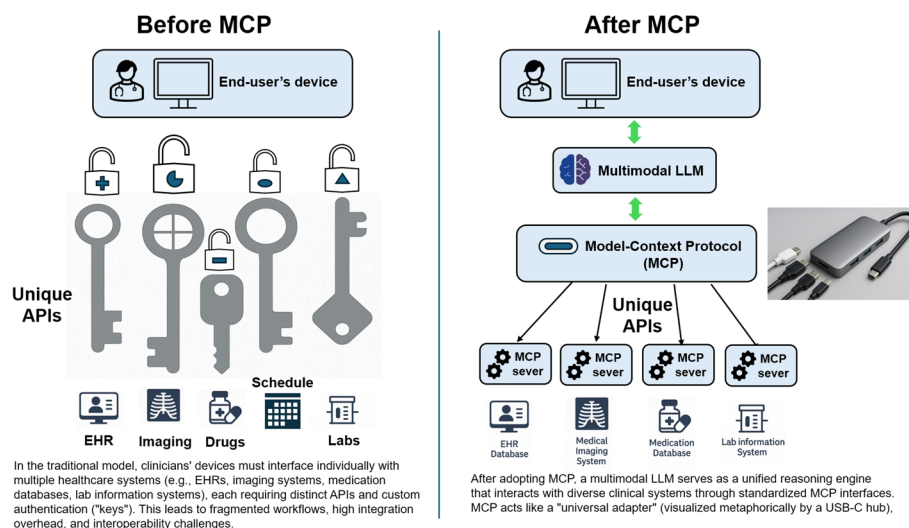


Fig. 3 Transformation of clinical system integration before and after model-context protocol (MCP) adoption. Before MCP, end-user devices had to interface separately with each clinical system (e.g., EHRs, imaging systems, medication databases, laboratory systems) via unique APIs and customized authentication ("keys"), leading to fragmented workflows and high integration burdens. After MCP adoption, a multimodal LLM interacts with diverse clinical systems through standardized MCP interfaces, with MCP functioning as a "universal adapter" (illustrated by a USB-C hub) that simplifies connectivity, enhances interoperability, and streamlines clinical operations

3.2 Outlook and strategic applications

The maturation of agentic systems is likely to shape the next phase of clinical AI. Rather than serving solely as question–answer tools, agents will progressively act as proactive collaborators that retrieve imaging studies, synthesize longitudinal EHR data, validate diagnostic criteria, cross-reference clinical guidelines, and draft documentation or orders with limited clinician input (Fig. 4). A practical near-term objective is the creation of safe “agent loops” in clinical settings, in which modular, role-specific agents (for example, triage assistants, documentation agents, guideline navigators) operate semi-independently under human supervision. The intended outcomes are increased clinical throughput, reduced administrative burden, and improved access to specialty care across diverse environments.

Table 3 outlines priority domains for agentic AI, including scalable interoperability across heterogeneous hospital systems, enhanced decision support via multimodal synthesis, automation of documentation and care transitions, continuous monitoring using wearable and IoT streams, expanded telemedicine through multilingual agents, acceleration of biomedical research and drug discovery, and personalized patient engagement through AI health companions. As an illustrative use case, during on-call hours an MCP-enabled agent could aggregate a patient’s evolving symptoms, imaging reports, and laboratory trends into a concise, auditable summary to support timely clinical decisions. The MCP is positioned as an enabling layer by standardizing secure, real-time access to clinical data, orchestrating tool-augmented reasoning, and supporting regulatory compliance. Realizing these benefits requires concurrent progress in model calibration and accuracy, bias mitigation, certification, privacy protection, and governance.

API, application programming interface; EHR, electronic health record; GDPR, General Data Protection Regulation; HIPAA, Health Insurance Portability and Accountability

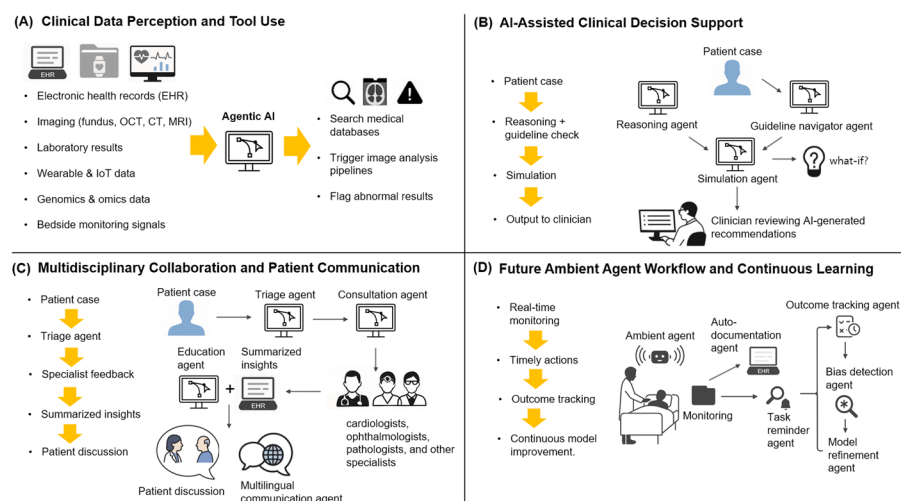


Fig. 4 Example scenarios of AI agents in clinical medicine. **(A)** Clinical Data Perception and Tool Use: Agentic AI ingests EHR, imaging, labs, wearables, omics, and monitoring data to search databases, run image analysis, and flag abnormalities. **(B)** AI-Assisted Clinical Decision Support: Patient cases pass through reasoning, guideline navigation, and simulation agents to generate “what-if” analyses and recommendations for clinician review. **(C)** Multidisciplinary Collaboration and Patient Communication: Triage and consultation agents coordinate specialist input. Education and multilingual agents deliver simplified, translated information for patient discussions and EHR updates. **(D)** Future Ambient Agent Workflow and Continuous Learning: Ambient agents monitor care settings, automate documentation, and send task reminders. Outcome tracking informs bias detection and model refinement for continuous AI improvement

Table 3 Future perspectives reshaping medicine with AI agents via model-context protocol (MCP)

Future perspective area	Main advancement	MCP's future roles
Scalable integration through MCP	MCP enables standardized, modular, and secure integration of AI systems with diverse clinical data sources, supporting real-time contextual reasoning	Acts as the secure connector between AI models and heterogeneous hospital IT systems (EHRs, LIS, PACS, scheduling) with standardized API endpoints
Enhanced clinical decision support and diagnosis	Future multimodal models will unify imaging, pathology, genomics, and clinical narratives into a single diagnostic framework for precision medicine	Provides real-time access to multimodal clinical data, enabling AI to synthesize imaging, labs, and genomics dynamically during reasoning
Revolutionizing clinical workflows	Agentic AI will automate documentation, care transitions, and multi-step workflow orchestration, minimizing administrative burden and clinician cognitive load	Facilitates AI-initiated workflows such as test ordering, follow-up scheduling, and discharge planning through tool-enabled action execution
Continuous patient monitoring and early intervention	AI agents will monitor wearable and IoT data streams continuously, interpret anomalies in real-time, and initiate early interventions via integrated alerts	Supports dynamic data retrieval from personal devices (wearables, home monitors) and hospital records for proactive, continuous patient care
Global healthcare access and telemedicine	Multilingual agentic AI systems will facilitate accessible triage, virtual consultations, and patient management in regions with healthcare workforce shortages.	Ensures multilingual AI agents can securely fetch patient histories, lab results, and scheduling data for global triage and telemedicine coordination.
Research and drug discovery	Agentic AI will enhance literature mining, therapeutic hypothesis generation, patient-trial matching, and molecular drug design using multimodal biomedical data	Allows secure retrieval of large-scale de-identified biomedical datasets, research protocols, and patient registries for AI-driven discovery
Personalized medical AI for consumers	Persistent AI health companions will support daily decision-making, preventive health, and personalized engagement by securely accessing real-time clinical data via MCP	Enables secure, role-based retrieval of patient-specific data (labs, prescriptions, history) for individualized coaching, alerts, and education
Challenges and the path forward	Addressing AI model bias, ensuring clinical-grade accuracy, establishing certification frameworks, and implementing robust privacy and safety governance are essential	Facilitates secure access control, encryption, auditing, and trust management between AI agents and clinical systems, ensuring compliance with HIPAA/GDPR
Ambient agentic AI	Always-on clinical assistants embedded in environments (e.g., exam rooms, wards) to anticipate needs and support workflows passively	Background access to real-time clinical signals, environmental context, and multi-agent coordination within institutional policy boundaries

API, application programming interface; EHR, electronic health record; GDPR, General Data Protection Regulation; HIPAA, Health Insurance Portability and Accountability Act; IoT, internet of things; LIS, laboratory information system; LLM, large language model; MLLM, multimodal large language model; PACS, Picture Archiving and Communication System

Act; IoT, internet of things; LIS, laboratory information system; LLM, large language model; MLLM, multimodal large language model; PACS, Picture Archiving and Communication System.

Although MLLMs can facilitate flexible data exchange, reliable interaction with heterogeneous systems will require agentic AI to rely on standardized data exchange mechanisms. MCP provides a mechanism for real-time interaction across clinical environments, and it can be aligned with established interoperability standards such as Fast Healthcare Interoperability Resources (FHIR) [72, 73]. FHIR is widely used to structure and exchange health data, including EHR content, laboratory results, and imaging meta-data. Integrating MCP with FHIR-based APIs could offer a scalable route to deployment across sites while maintaining institutional controls and compliance with regulatory frameworks.

The next stage is likely to move beyond prompt-driven assistants toward ambient agents that operate continuously within care environments. Rather than waiting for

explicit queries, agents may interpret context from clinician–patient conversations, EHR activity, and bedside device signals, and then provide timely, unobtrusive support within routine workflows. In hospitals, such agents could draft documentation, surface guideline passages during diagnostic reasoning, or flag overlooked tasks, while interacting securely with institutional systems through protocols such as MCP. If implemented with clear oversight and auditability, these capabilities have the potential to reduce cognitive burden and strengthen care coordination.

3.3 Limitations of agentic AI via MCP

While the integration of agentic AI systems with clinical environments through MCP-style integration offers transformative potential, several limitations must be critically addressed before widespread adoption is feasible [5]. These challenges span technological immaturity, unresolved legal and ethical considerations, and practical operational barriers such as computational cost and latency.

- **Tool and protocol immaturity:** Although MCP offers a promising framework for standardized and modular integration of AI agents with healthcare systems, its current ecosystem remains largely experimental. Production-ready deployments in clinical environments are rare, and adoption among hospitals and EHR vendors is limited. Interoperability often requires custom middleware adaptations rather than true plug-and-play solutions. Furthermore, standardized MCP interfaces for essential clinical functions, such as drug interaction checking, diagnostic-guideline retrieval, and laboratory data access, are underdeveloped. These limitations restrict the scalability and operational efficiency of agentic AI systems. Broader adoption will require extensive vendor collaboration, regulatory endorsement, and the development of robust open-source tooling. Future implementations of MCP-enabled AI agents could leverage FHIR-based data exchange to enhance compatibility with existing health infrastructure and facilitate regulatory approval in real-world deployments [74].
- **Security, authentication, and authorization gaps:** MCP currently lacks a unified framework for identity verification, access control, and secure communication between clients and servers. Without standardized authentication protocols and permission management, MCP-based systems are vulnerable to unauthorized tool access, privilege escalation, and data leakage, especially in multi-tenant environments. Additionally, the absence of centralized logging, monitoring, and security validation mechanisms impairs the ability to detect, trace, or mitigate malicious behavior. These gaps present serious risks in clinical settings, where system integrity and patient data confidentiality are paramount [75]. Addressing them will require the development of secure-by-design standards, mutual authentication schemes, and auditable access control layers tailored to agentic AI infrastructures.
- **Legal, ethical, and liability challenges:** The evolution of AI from passive decision-support tools to semi-autonomous agents introduces complex legal and ethical challenges. Current regulatory frameworks, including those established by the FDA, HIPAA, and GDPR, do not fully address scenarios where AI agents autonomously initiate clinical actions or decisions through MCP-mediated interactions. Issues of accountability are particularly problematic. If an AI agent issues an erroneous

recommendation, it is unclear whether liability would fall on the healthcare provider, the AI developer, or the integration vendor. Furthermore, the opacity of LLMs complicates informed consent and explainability, as patients may not realize that decisions are influenced by AI systems. Addressing these concerns will require the establishment of governance models, shared responsibility standards, and transparent auditing frameworks tailored to agentic AI applications.

- **Computational cost and latency:** MLLMs with agentic capabilities impose substantial computational burdens, posing challenges for real-time clinical integration. The inference demands of MLLMs, especially when processing multimodal data streams such as imaging, laboratory results, and clinical narratives, can introduce delays that are incompatible with time-sensitive clinical workflows. In low-resource or mobile-first environments, these computational requirements further limit deployment feasibility. Moreover, sequential tool invocations and multi-server MCP queries may accumulate latency, disrupting clinician efficiency. Addressing these limitations will require advances in model optimization techniques, including quantization and distillation, deployment of lightweight edge AI systems, and strategic prioritization of time-sensitive inference tasks. It will also require the development of efficient methods for compressing healthcare data and improving its transmission [76].

3.4 Proposed checklist for safe deployment

Table 4 proposes an idealized security and privacy checklist for MCP-style integration deployment in healthcare, synthesizing best practices from AI safety, clinical informatics, and regulatory compliance frameworks. These recommendations are intended to guide future MCP implementations, with prioritization adapted to the specific risk profile and resource constraints of each healthcare environment. This framework categorizes essential safeguards across access control, data security, context management, tool governance, auditing, regulatory compliance, system architecture, and user education. Key recommendations include enforcing robust identity and session management, ensuring encrypted and tamper-evident exchanges, isolating patient-specific context to prevent cross-contamination, restricting agentic tool actions with human oversight, and establishing comprehensive governance and auditing processes. Adoption of these practices will be critical for ensuring that MCP-enabled agentic AI systems operate securely, ethically, and in compliance with healthcare regulations while maintaining clinical trust and reliability.

It should be emphasized that the recommendations summarized in Table 4 represent an ideal target framework rather than a rigid checklist for immediate implementation. Given the current immaturity of MCP ecosystems and variability in institutional resources, not all security, interoperability, and governance measures can be simultaneously achieved in early real-world deployments. Nonetheless, this framework provides a strategic vision toward which MCP-based clinical AI integrations should progressively evolve, balancing technical feasibility with regulatory compliance and clinical practicality.

Table 4 Proposed best practices for secure and privacy-preserving MCP-style integration deployment in healthcare (ideal target framework)

Classification	Item number	Recommendation	Healthcare example
Access control & identity management	1	Implement mutual authentication between MCP clients and servers (e.g., via OAuth 2.0, API tokens, or mTLS)	Prevents unauthorized AI tools from retrieving EHR data
	2	Use role-based access control to restrict which users/agents can access specific MCP endpoints	Restricts a documentation bot from accessing lab ordering functions
	3	Assign unique client IDs for every MCP client, and log usage per client	Identifies if a specific diagnostic agent is over-querying a system
	4	Enforce session expiration and token rotation	Invalidates stale tokens after a telehealth consultation ends
Data security	5	Ensure all MCP communication uses TLS 1.2+ encryption (HTTPS)	Prevents PHI leakage during AI-driven image retrieval
	6	Apply encryption at rest for any cached or retrieved data (e.g., temporary result storage)	Secures AI-generated discharge summaries awaiting review
	7	Sanitize all inputs and outputs to prevent injection vulnerabilities or prompt leakage	Stops injection attempts in AI-driven order entry
Context handling	8	Limit the duration and scope of retained context in memory (especially patient data)	Erases patient identifiers after completing a diagnostic task
	9	Tag and segregate PHI from non-sensitive data in model memory buffers	Shares vital signs without revealing patient name
	10	Do not allow cross-patient context contamination in multi-session or multi-threaded AI tools	Prevents AI from mixing lab results from two patients in a report
Tool safety (agentic function control)	11	Maintain whitelist of callable MCP tools/functions with risk classification	Allows read-only access to imaging archives, blocks prescription writing
	12	Require human approval before executing any agent-triggered action (e.g., medication orders, referrals)	Doctors must approve AI-suggested surgery orders
	13	Rate-limit high-risk operations to avoid AI-triggered overloads or denial-of-service scenarios	Limits AI-driven mass lab orders during system errors
Monitoring, logging, and auditing	14	Log every MCP call: client ID, timestamp, server endpoint, payload summary (without full PHI)	Tracks which agents accessed oncology reports
	15	Regularly audit logs for unauthorized or anomalous behavior	Identifies repeated failed attempts to access imaging servers
	16	Use tamper-proof logging systems (e.g., append-only logs with cryptographic hash chaining)	Maintains chain-of-custody for AI usage review
Regulatory compliance	17	Ensure MCP deployment aligns with HIPAA, GDPR, HITECH, or other local data privacy laws	Meets GDPR requirements for EU telemedicine patients
	18	Sign Business Associate Agreements with AI and MCP service providers if PHI is handled	Formalizes compliance for outsourced AI transcription
	19	Maintain clear data lineage documentation: what flows where, under what protections	Shows exactly where patient imaging data is routed
System design & isolation	20	Deploy MCP servers in isolated, secured subnets.	Hosts MCP orchestration node inside hospital firewall.
	21	Avoid public internet exposure unless gated by API proxy/WAF	Requires VPN for AI-agent hospital access
	22	Segment access between internal (hospital) and external (third-party) data systems via proxy layers	Allows AI to retrieve drug prices from public APIs but not submit patient data

Table 4 (continued)

Classification	Item number	Recommendation	Healthcare example
User education & governance	23	Train staff on AI safety boundaries, especially regarding overreliance on autonomous outputs	Educates nurses on validating AI triage suggestions
	24	Define clear policies for responsibility and override authority in human-AI collaboration	Clarifies that physician judgment supersedes AI treatment plans
	25	Establish an AI Governance Committee to review risks and outcomes of MCP-integrated tools	Periodic review of AI-driven screening tool outcomes

API, application programming interface; EHR, electronic health record; GDPR, General Data Protection Regulation; HIPAA, Health Insurance Portability and Accountability Act; HITECH, Health Information Technology for Economic and Clinical Health Act; mTLS, mutual transport layer security; MCP, model-context protocol; PHI, protected health information; TLS, transport layer security; VPN, virtual private network; WAF, web application firewall

3.5 Obstacles to the responsible adoption of agentic AI in healthcare

The promise of agentic AI in healthcare is substantial, yet clinical deployment should proceed cautiously with attention to operational, regulatory, and ethical requirements. A central question is the appropriate level of autonomy. Current evidence favors semi-agentic systems that assist clinicians in documentation, administrative triage, and step-wise diagnostic reasoning, rather than replacing clinical judgment [77]. Full autonomy for high-stakes decisions is ethically contentious and operationally unproven without continuous human oversight. A risk-stratified oversight model is therefore pragmatic: low-stakes functions such as logistics or summarization may allow greater automation; moderate-risk activities such as insurance preauthorization or initial triage should require mandatory supervisory review; high-stakes actions, including initiation or modification of treatment, should remain under direct human control, with AI limited to an assistive role [78].

Secure, interoperable access to real-time health information is equally critical. Security and privacy are foundational prerequisites, not ancillary technical details. Prior work has described a remote-to-device protocol that enables retrieval of encrypted, cloud-based EHR backups during emergencies, granting access to authorized clinicians through dynamic trust validation and time-limited tokens [79, 80]. This demonstrates that real-time, privacy-preserving exchange is feasible while maintaining consent and data sovereignty. Agentic AI should operate within strict access-control frameworks that ensure data are used only in the intended clinical context. Future implementations of the MCP will require comparably robust controls, including role-based access, centralized logging, and dynamic consent, to meet privacy obligations while preserving operational responsiveness. Safe deployment depends not only on model performance but also on the integrity, auditability, and governance of data interactions.

MCP-based implementation alone does not guarantee safety or effectiveness. Real-world use requires standardized mechanisms for identity management, auditability, and fail-safe behavior, particularly in multi-agent settings. Regulators such as the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA) will need to define guidance for orchestration, liability, and transparency in multi-step, tool-invoking workflows. The research community should in parallel advance external validation across diverse sites, develop metrics for multi-step reasoning and agentic tool invocation, and create simulation environments that reflect operational conditions, including

noisy or incomplete inputs. Without progress on these fronts, agentic AI is likely to remain promising yet fragmented rather than a trusted component of clinical care.

3.6 Open research questions

Future work should build a science of reliable, accountable clinical agents. Priorities include prospective, longitudinal evaluations that bound error propagation and communicate calibrated uncertainty for multimodal outputs. Privacy-by-design methods are needed, such as selective retrieval, federated or split inference, and differential privacy [81]. Security should adopt shared red-teaming protocols and benchmarks that cover prompt injection, tool misuse, supply-chain risks, and multimodal adversarial attacks that combine text, images, and structured data [82]. Governance must define practical accountability and required audit artifacts, while equity research develops scalable subgroup testing and mitigation across languages, devices, and sites. Interoperability should align MCP calls with FHIR and add end-to-end observability, and engineering studies should meet bedside latency through local preprocessing, caching, and model distillation alongside human-factors trials, silent deployments, and continuous post-market monitoring.

4 Discussion

Early attempts to integrate AI into healthcare were limited by bespoke, point-to-point links between models and data sources, narrow support for heterogeneous modalities, and the absence of standardized communication protocols. These ad hoc integrations were difficult to generalize, expensive to maintain, and fragile as clinical systems evolved. Tooling ecosystems were immature, and interoperability gaps prevented routine use beyond narrow pilots. A core lesson from this history is the need for a universal, standards-aligned integration layer that provides secure, dynamic connectivity across clinical systems; the MCP is a candidate framework for this role.

Our perspective rests on several working assumptions. First, capabilities of multimodal LLMs and agentic AI architectures will continue to advance. Second, adoption of standards and protocols such as MCP, ideally aligned with existing healthcare standards, will grow among vendors and institutions to address interoperability and security. Third, AI will shift from passive prediction to goal-directed assistance that augments, rather than replaces, clinical judgment. At the same time, technical progress alone is insufficient. The practical value of MCP or any agentic framework depends on vendor collaboration, regulatory guidance, and robust open-source and commercial tooling. Human-in-the-loop oversight remains essential, particularly for high-stakes decisions, to ensure safety, accountability, and alignment with clinical intent.

This work has limitations. MCP remains early in its lifecycle, and production-grade clinical deployments are scarce; consequently, many claims are therefore conceptual rather than empirically validated. The scalability, reliability, and operational efficiency of MCP-mediated agent workflows in complex hospital environments require prospective evaluations across sites and services. In addition, issues of calibration, bias, and distribution shift persist for current models and must be addressed alongside integration concerns.

Potential beneficiaries span the healthcare ecosystem. Clinicians may see reduced administrative burden and improved access to contextual information at the point

of care. Health-system leaders may realize operational efficiencies through automation of routine tasks (for example, authorization, claims, and logistics). Developers and researchers gain a clear architectural target for interoperable, auditable systems and a research agenda that emphasizes standardized protocols, observability, and safety controls. Patients may benefit from better care coordination, timelier decisions, and clearer communication. Dissemination should target both clinical and informatics audiences through conference presentations, journal publications, and contributions to shared tooling and specifications.

Looking ahead, we envisage a phased roadmap. In the near-term (3 to 4 years), priorities include building realistic simulation and silent-mode evaluation environments for multi-step reasoning and tool invocation, and piloting MCP alignment with existing standards such as FHIR to ensure compatibility with incumbent infrastructure. In the mid-term (4 to 7 years), efforts should focus on collaboration with regulators (for example, the U.S. Food and Drug Administration and the European Medicines Agency) to develop guidance on orchestration, liability, auditability, and transparency in multi-agent workflows, alongside controlled pilots that quantify clinical utility, safety, and operational impact using predefined metrics. In the long-term (7 years and beyond), the goal is ambient, MCP-mediated agentic assistance embedded in routine workflows, supported by mature security and governance, including role-based access, centralized logging, dynamic consent, and clearly defined fail-safe behavior. Progress along this path requires coordinated advances in model reliability, integration standards, safety engineering, and governance. Without such coordination, agentic AI is likely to remain promising but fragmented. With such coordination there is a credible route to trustworthy, auditable assistance that enhances clinical care.

5 Conclusion

The convergence of MLLMs, agentic systems, and MCP-style integration positions AI to act as a supervised, goal-directed collaborator rather than a passive analytic tool. When coupled with interoperable access to EHRs, imaging, and laboratory systems, agents can retrieve context, reason over longitudinal data, and assist with documentation and coordination under explicit human oversight. The near-term opportunity is not autonomy but dependable assistance that reduces cognitive and administrative burden and improves the timeliness and consistency of care.

Realizing this potential requires progress on several fronts. Models must demonstrate calibration, interpretability, and robustness across sites, services, and patient subgroups. Interoperability must be standardized, with secure, auditable data exchange that aligns with institutional policy and regulation. Governance should define clear boundaries of use, identity and access controls, auditability, fail-safe behavior, and pathways for evaluation including non-interventional live evaluation, latency targets, override metrics, and equity analyses. Legal and ethical frameworks must evolve to address accountability and transparency in multi-step, tool-invoking workflows.

A measured path forward combines technical innovation with careful integration in clinical operations and sustained attention to safety and equity. This perspective outlines the architectural principles and evaluation practices needed to support that path. With interdisciplinary collaboration, iterative validation, and a patient-centered focus, agentic

AI integrated through standards such as MCP can mature into trustworthy, auditable assistance that enhances clinical care.

Acknowledgements

Use of generative AI – The authors used generative language tools (for example, Microsoft Copilot and standard grammar checkers) only for copyediting and formatting. No tool generated scientific ideas, claims, analyses, figures, or tables. The authors wrote and verified all substantive content, selected and checked all references, and take full responsibility for the manuscript. All AI-assisted edits were reviewed by the authors for accuracy, appropriateness, and consistency with journal policies.

Author contributions

Joon Yul Choi: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. Tae Keun Yoo: Conceptualization, Project administration, Resources, Supervision, Writing – review & editing.

Funding

None.

Data availability

Not applicable. No original datasets were generated or analyzed during the preparation of this manuscript. All information discussed is based on publicly available literature and frameworks.

Declarations

Ethics approval and consent to participate

Not applicable. This study is a conceptual and technical perspective that does not involve human subjects, animal experiments, or the use of identifiable personal data. Not applicable.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form, including images or clinical details.

Competing interests

The authors declare no competing interests.

Received: 27 April 2025 / Accepted: 21 January 2026

Published online: 24 January 2026

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